

A Soliton Cellular Automaton

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Source: Journal of the Physical Society of Japan, Vol. 59, No. 10, pp. 3514–3519 (1990)

ABSTRACT

We propose a 1-dimensional discrete cellular automaton consisting of an infinite array of boxes and a finite number of balls. Despite being governed by simple boolean transition rules without differential equations, the system exhibits exact soliton phenomena: localized pulses of consecutive balls retain their identity and velocities through mutual non-destructive phase-shifted collisions. Under the Takahashi-Satsuma rule, larger solitons travel strictly faster than smaller solitons, naturally decomposing an arbitrary initial configuration into a spatially sorted train of solitons.

1. The Ball-and-Box Rules

Consider an infinite 1D array of boxes indexed by integers j in $\mathbb{Z}$, where each box is either empty (0) or holds one ball (1). The total number of balls is finite.

The evolution from time t to $t+1$ is executed via a 'carrier' mechanism:

1. Start from the far left where all boxes are empty.
2. Move to the right box by box.
3. If the box contains a ball, pick it up (carrier load increases by 1) and empty the box.
4. If the box is empty and the carrier holds balls, drop one ball into the box.
5. Repeat until all balls in the carrier are deposited.

2. Soliton Solutions & Velocity Formula

A block of m consecutive balls (surrounded by empty boxes) behaves as a solitary wave of amplitude m .

Under the carrier rule, a solitary wave of length m advances exactly m boxes in one time step: $v(m) = m$.

Because velocity is proportional to size, larger solitons travel faster than smaller solitons. When a large soliton overtakes a small soliton, they undergo an elastic collision with an exact spatial phase shift: $\Delta x = 2 * \min(m_1, m_2)$.

3. Exact Integrability and Sorting Property

The Box-Ball System (BBS) is the ultradiscrete limit (tropicalization) of the Korteweg-de Vries (KdV) and Toda lattice equations via the substitution $\lim_{\epsilon \rightarrow 0} \epsilon \ln(e^{A/\epsilon} + e^{B/\epsilon}) = \max(A, B)$.

Any initial disordered configuration of balls decomposes under BBS dynamics into a sorted train of non-interacting solitons arranged from right to left in order of descending speed (or left to right in ascending speed).